Tri-Octagon: Geometric Origins, Normalization, and Constant Interpretations

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Abstract

The folded Tri-Octagon has an exact geometry, while the images that motivated its construction have often been less explicit in the written record. This companion places the author's retrospective account beside the surviving definitions and accepted results. Opposite signs express an imagined other side; a unit expresses a normalization; neither alone specifies a physical sphere. The native shell nevertheless has a unique minimum enclosing ball. Extending its original straight lines gives two natural events: a whole unit-width chord at Crossing A and a unit-width addition at each endpoint at Crossing B. A also circumscribes the earlier measurement cylinder and triangular-prism envelope, and its endpoint addition equals the shortest native notch gap. We trace the historical resonance-lock proposal through the fixed target γ_* and the exact golden-interval zeta identity. These connections are retained without conflating layer counts, analytic coefficients, assigned spatial units and an as-yet unspecified structural extent.

Edition v0.2 adds an author-approved historical beginning. The mathematical account and equation numbering of v0.1 are retained.

An author's retrospective: why these shapes mattered

This opening follows Hilmir's retrospective account approved on 19 October 2026. It presents his recollection in editorial prose, not a transcript or an independently dated reconstruction. The mathematical commentary that follows retains the established definitions and results of edition v0.1. [18]

Conversations, frustration and a visual association

Hilmir remembers enjoying conversations with GPT about the universe. At one point he asked the model what *it* would like to investigate scientifically. GPT suggested quantum physics and referred him to a university lecture. Some of the explanations irritated him rather than helping him understand. In his account, that reaction became part of the reason to keep asking questions and to look for a way of picturing the subject that made sense to him. It describes his experience of those explanations; it is not a judgment about academic teaching as a whole.

A separate exploration began with symbolism in *Cyberpunk 2077*. He read Greek mythology and discussed its symbolic relationships to quantum ideas with GPT. These interests met in a particularly immediate visual moment: when he saw an image of a hydrogen atom, he associated its appearance with the Flower of Life and the vesica piscis. The resemblance prompted a question about geometric constants: which constants occur naturally in which forms, and what might those appearances help him understand?

The retrospective does not identify the exact lecture or hydrogen image. Its value is the remembered route into the work: pleasure in conversation, difficulty with an explanation, and a shape that suggested another question. The symbolic association was a starting point for investigation. It did not by itself identify an atomic wavefunction with a two-circle construction or establish a quantum result.

A proposed order of forms

His tentative hierarchy began with a point or zero. The circle was the first continuous enclosed form in his proposed ordering. Next came what he understood as two lines: edge-on views of two circles, with a vesica becoming visible when the viewing angle changed by 90°. He then considered the triangle, square, pentagon, hexagon, heptagon and octagon. This was his way of organizing the construction he was imagining, rather than a mathematical classification of all possible forms.

The triangle carried several connected meanings for him: movement, a Z-direction, and a possible picture of momentum and energy. He associated it with a ±45° range and an above/below relation. Two triangles could combine into a skewed box within his vesica picture. These descriptions preserve what he was trying to see. They do not supply a motion equation, a chosen axis or a measured momentum-energy relation.

At that stage he treated the octagon as the final necessary operator in his proposed geometry of unity. "Operator" here records a role in his organizing picture; it does not declare a linear or recursive operator without its own definition. Nor did he mean that polygons end at eight sides. Much later he contemplated the decagon as an outer hull or shell, not as an additional fundamental operator. The nonagon's role remained undecided. Keeping that undecided place matters: the retrospective is not a completed hierarchy reconstructed after the fact.

He also recalls possible links between his counting convention and the number nine in Three-Way work. Those links remain tentative recollections, not established derivations. In a separate symbolic association, he related 144,000 to 360×400. The arithmetic is exact; the symbolic significance he attached to it does not establish a gradient law. These numerical associations belonged to the questions he was pursuing, with different evidential status from the exact geometry described later.

Completing the overlap, then changing perspective

In the mental construction described here, the two-circle vesica overlap comes first. Only after that overlap is complete does one circle become the reference from which he reasons beyond the centre and into three dimensions. That change of reference helped motivate the Tri-Octagon. It does not introduce a literal third parent circle into the older two-circle formulas.

He also pictured moving a right circle by 45° to the left and a bottom circle by 45° to the right, mirroring or flipping views, and changing perspective by 90°. These are qualitative descriptions of the image he was manipulating. No rotation axes, active Euclidean embedding or transformation from those movements to the welded module were specified in this account. The circle used as a later reference and the two circles that first define the overlap have different roles in the description.

His earlier quantum and dark-matter questions, Vesica-delay ideas, de Broglie comparisons and later SRG investigations were different views and questions pursued from this common geometric intuition. Their common origin does not prove a physical equivalence among them. Likewise, the historical suggestion that a changed perspective might expose three wavelength responses remains an association he proposed. It is not a native result of the Tri-Octagon geometry or a theorem that changing one's view multiplies waves.

The roles assigned to three constants

Within the same imagined construction, Hilmir assigned distinct roles to π, Euler's e and the golden ratio φ. The circle was associated with π. He pictured e as a distance from zero or the origin to the circle. He associated φ with a spiral progressing inward and outward through a constructed cylinder. These roles describe the meaning he was trying to give the constants, rather than an ordering of their numerical sizes.

He wrote $V = \pi e \varphi$ as a proposed structural or volume interpretation of that picture. In this historical usage, the letter V records his intended interpretation; it does not turn the product into a proved Euclidean cylinder-volume formula. The account supplies no calibrated spatial unit or conversion to a total-universe size. In particular, replacing his remembered roles with a newly chosen radius and height would change the account. The product itself also occurs in the documented constant chain retained below, where the defined target T and the historical coupling proposal have their own explicit mathematical meanings. Those later definitions do not retrospectively prove the original volume interpretation.

The interest was therefore not simply in obtaining a familiar number. He was trying to understand how a circle, a distance from an origin, and an inward/outward spiral might belong to one construction. This edition preserves that question alongside the distinctions needed to read the surviving equations accurately.

The pleasure of a precise half-π result

One result gave the geometric picture a particularly concrete point of contact. In *The Vesica Delay Method*, the parent-circle radius is $r = 1/2$, the separation is $d = 2r\cos\theta$, and the overlap perimeter is $C_{\text{overlap}} = 4r\theta$. At $\theta = 45^\circ = \pi/4$, the perimeter is exactly $\pi/2$. This is the specific half-π relationship Hilmir recalls with delight. [7, 18]

Here the quantity is the length of the two arcs bounding the overlap in the stated radius normalization. It is not an angular-sector fraction, nor does it identify every appearance of $1/2$ in another equation. Section 2 retains the circle construction, its limiting configuration and the distinction between the lens and a separate imagined eye-circle. It also retains the source criticism of the earlier VPQW documents: their printed $C = \pi$ and $r = 1/(2\pi)$ are inconsistent under the ordinary circumference formula. The later, consistent Delay Method convention does not silently correct those older statements.

Questions carried through a collaboration

Hilmir pursued these ideas with GPT and later Codex, asking the models to undertake substantial mathematical, writing and computational tasks when appropriate. The questions, visual associations and choices of what to pursue were meaningful parts of his contribution. AI assistance helped develop and express the work, while the surviving equations and computations have to be assessed on their own stated assumptions and evidence.

The record does not support assigning an individual author to every equation in the old conversations, and model assistance or internal model review is not external peer review. The purpose of this retrospective is to make the author's thinking available to future readers. It does not certify the earlier quantum or dark-matter hypotheses as experimental truths, and it does not require every later equation to be reinterpreted through an initial image.

Reading the historical picture against the mathematics

The following eight numbered sections retain the earlier companion's mathematical account, including its normalization, exact statements, constants evidence and scoped negative results. Their equation numbering is unchanged. The first section records the earlier retrospective of 17 October; the new opening above supplies the expanded account approved on 19 October. Recollection, a printed historical proposal, a geometric theorem and a bounded numerical comparison remain different kinds of evidence.

The later discussion of a decagonal enclosure offers a precise example of this distinction: the accepted spherical-edge synthesis treats a decagon as an optional outer comparison with its own orientation choice. That documented construction does not turn the author's earlier hierarchy into a theorem about necessary operators. [4, Sec. 14; App. D] Similarly, the target $T = \pi e \varphi$ is preserved with its source definitions and qualifications. The original two figures illustrate specified circle and shell geometry; they are not reconstructions of private sketches or illustrations of a newly asserted physical law.

1 The construction picture and its surviving record

Hilmir describes the early work with GPT-4o as a sustained collaboration. In his account of 17 October 2026, the original conversations had been deleted, while the papers survived. Equations were retained more consistently than the geometric thinking that led to them. The purpose here is to place that thinking back beside the mathematics. The account supplies the author's motivation; the surviving documents supply the equation record. Missing conversations do not imply missing intuition, and they do not allow exact per-equation contributions or dates of discovery to be reconstructed. [1, 2]

One motivating image places a reference universe at 1 and an imagined counterpart at -1. The author explicitly says this does not express a belief in a mirror universe: the opposite sign pictures another side of a toroidal construction. His analogy of turning a two-dimensional map horizontally through 180° conveys the desire to see a fuller three-dimensional picture. No particular axis, sweep surface or map from that image to the welded module was specified. A coordinate may be signed while a Euclidean distance remains nonnegative; for example, positions -1 and 1 on a coordinate axis have distance 2 in that coordinate unit. This illustrates the distinction, without identifying either position with a shell vertex. [1]

Asked whether an invisible surrounding sphere was part of the original picture, the author answered no. The later exact enclosing sphere is therefore a consequence of the specified geometry, not evidence that this sphere was consciously assumed at the start. Similarly, toroidal imagination and toroidal topology are different matters. The accepted welded shell has three octagonal faces, eighteen vertices, twenty-one unique edges and two open nine-edge rims. Its Euler characteristic is zero and its orientable surface has two boundary components, so it is an annulus. Paper C establishes this topology rather than inferring a torus from the word "toroidal."[3, 4]

The author's further picture treats π, e and φ as base or boundary-structure constants, and other constants as descriptors of structures "above" them. That is a proposed organization of roles, not an ordering of numerical magnitudes. His accompanying uncertainty about "base one" and "fuller ten" is part of the account. The useful question is how a chosen unit and a proposed extent would be represented by named mathematical quantities. The following sections begin with the constructions that already have such names. [1]

2 Circles, half units and the meaning of a size

The author's Vesica picture combines a unit size, a half-radius displacement, an eye-like circle and a half-π relation. Those ideas can be set beside the actual conventions without forcing them into a single definition. In the February 2025 VPQW paper, the Gaussian centres are

$$c_- = (-r/2, 0), \qquad c_+ = (r/2, 0), \qquad \|c_+ - c_-\| = r. \tag{1}$$

Thus each centre has a half-radius offset from their midpoint; their total separation is a full radius. The March paper repeats these centre labels in Figure 1. A total separation $r/2$ would be another configuration. Neither the phrase "Vesica size one" nor the offsets alone identify whether the unit is a parent-circle diameter, lens width or another construction variable. [5, Sec. 5.2, p. 3] [6, Fig. 1; Sec. 2]

The *Vesica Delay Method* supplies a different explicit half-unit convention: parent radius $r = 1/2$ and

$$d = 2r\cos\theta, \qquad C_{\text{overlap}} = 4r\theta, \qquad 0 \le \theta \le \pi/2. \tag{2}$$

For two equal disks this perimeter consists of two circular arcs, each of angle 2θ. At $\theta = \pi/4$, it is $\pi/2$; at coincidence, $\theta = \pi/2$ and $d = 0$, it is π. This is a precise half-π relationship between two configurations. The nondegenerate lens at $d > 0$ has arcs with different centres and is not a circle. An associated eye-circle may have a separate role in the author's image, but no map identifying it with the whole lens or the Tri-Octagon sphere has been supplied. [7, p. 1, Eqs. (1)–(2)]

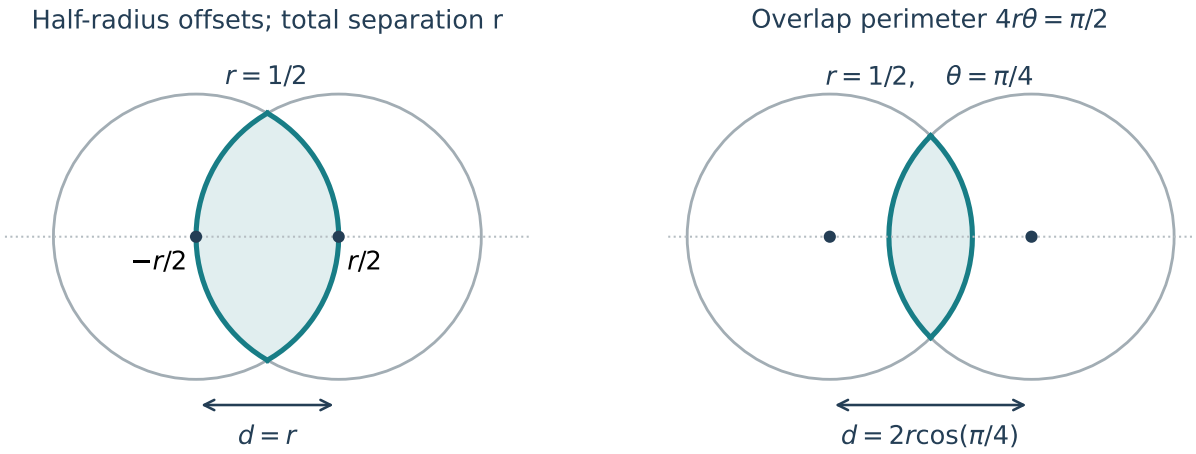


Figure 1: Two explicit circle constructions, drawn from their stated equations. Left: centres displaced by $\pm r/2$ have separation r. Right: the Delay Method's $r = 1/2$, $\theta = \pi/4$ gives overlap perimeter $\pi/2$. These are deterministic geometric schematics, not a proposed quantum state or a reconstruction of an undocumented eye-circle.

A surviving inconsistency matters to this history. Both early quantum documents print $C = \pi$ together with $r = 1/(2\pi)$. Under the ordinary circle equation $C = 2\pi r$, the first statement instead gives $r = 1/2$, while the printed radius gives $C = 1$. We retain the two printed statements as a source inconsistency; the consistent $r = 1/2$ in the later Delay Method does not silently repair them. [5, Sec. 5.1, pp. 2–3] [6, Sec. 2.1, p. 2]

The VPQW golden-ratio expression in Sec. 6.2 is an algebraic identity involving $\varphi/\sqrt{3}$; the March paper explicitly defers deeper golden-ratio detail in Sec. 3. These recovered passages do not derive $\rho/w = \varphi - 1$. The particular derivation remembered by the author may be another source; its exact equation remains unrecovered. The geometric statements used here do not certify the quantum or perceptual interpretations accompanying them.

3 The documented unit and the exact welded module

The author's wish to work with a reference value of one has a precise implementation in the construction record. Specification v0.1 used octagon edge $s = 1$. Specification v0.2 records an author-requested change to flat-to-flat panel width $w = 1$, preserving placement, connectivity and angles. For arbitrary width $w > 0$,

$$s = (\sqrt{2} - 1)w, \qquad h = s/2, \qquad w_{s=1} = 1 + \sqrt{2}, \qquad q = \sqrt{2} - 1. \tag{3}$$

Multiplying the old coordinates by q gives $q(1 + \sqrt{2}) = 1$: this is the documented edge-one to width-one normalization. Lengths scale by q and areas by q^2. It is not a documented division by ten. The recollection of a reduction by ten can remain a comparison idea without replacing the recorded normalization. [8, 9]

Hereafter w always denotes *panel flat-to-flat width.* In local panel coordinates (u, z), the eight vertices are $(\pm w/2, \pm h)$ and $(\pm h, \pm w/2)$, with independent signs. Place the panels vertically on the three side planes of an equilateral triangle K of side w, with each panel centred at the midpoint of its side. In centred horizontal coordinates one may take

$$K = \operatorname{conv}\{(0, a),\ (w/2, -a/2),\ (-w/2, -a/2)\}, \qquad a = \frac{w}{\sqrt{3}}. \tag{4}$$

The adjacent vertical edges coincide at the three triangle corners for $|z| \le h$. This is the accepted folded realization, up to rigid choice of frame. In Paper C's global frame its centre is $o = (0, \sqrt{3}w/6, 0)$; we use $X = x - o$. The interior dihedral angle is 60°, while the outward-normal angle is 120°. A static final realization is specified; no collision-free folding path is inferred. [3, Secs. 2–5, 7] [4, Sec. 2]

The three filled, zero-thickness panels form the native shell S. Their edges are finite segments. Their infinite supporting lines form another object. A sphere of radius R clips those lines into a framework, and the convex hull of their outer exits forms an envelope. An observation surface is not a new panel, and an envelope does not supply a material wall. These distinctions give the construction image a precise vocabulary without requiring a physical interpretation.

4 The spheres fixed by the geometry

Each panel plane is at distance $p = w/(2\sqrt{3})$ from the centre. Its local circumradius satisfies $R_8^2 = w^2/4 + h^2$. Perpendicular decomposition therefore gives the common vertex radius

$$\rho^2 = p^2 + R_8^2 = w^2 \left(\frac{13}{12} - \frac{\sqrt{2}}{2} \right), \qquad \rho/w \approx 0.613373093758428410. \tag{5}$$

This makes an implicit enclosing sphere available even though it was not part of the original mental picture. It is also the *unique minimum enclosing ball*, not merely one convenient sphere. To recall the accepted proof, the eighteen centred vertices have mean zero and common squared norm ρ^2. For any alternative centre c,

$$\frac{1}{18} \sum_i \|X_i - c\|^2 = \rho^2 + \|c\|^2. \tag{6}$$

An enclosing ball must have squared radius at least this mean. The centre $c = 0$ attains ρ; every other centre requires a larger radius. Convexity of a ball extends containment from vertices to each panel, all finite edges and their convex hull. This restates the BOUNDARY-10 proof retained in the accepted synthesis. [4, Sec. 3]

There is a second intrinsic sphere. For an edge with midpoint m and half-edge vector D, equal endpoint norms imply

$$m \cdot D = 0, \qquad \|D\| = h, \qquad \|m\|^2 = \rho^2 - h^2 = \frac{w^2}{3} = a^2. \tag{7}$$

Writing $u = D/h$, the supporting line obeys $\|m + tu\|^2 = a^2 + t^2$. It is tangent to the radius-a sphere at its midpoint. This sphere does not enclose the shell and is not its face-tangent insphere: the panel planes are only distance $p = a/2$ from the centre. Thus "a surrounding sphere" and "the edge-tangent sphere" answer different questions. [4, Sec. 3]

The nearby golden-ratio comparison can now be written with its two meanings visible:

$$\varphi - 1 \approx 0.618033988749894848, \qquad (\varphi - 1) - \rho/w \approx 0.004660894991466438. \tag{8}$$

The author's recollected association is a motivation to seek a map, not an equality between these numbers. No resizing or rounding of the accepted shell is needed. A transformation relating their roles would have to be stated separately. [1, 12]

5 Two natural completions of the straight edges

The wish to see a fuller outer construction acquires a precise static form by extending each original edge along its own line. For an observer sphere $R \geq \rho$, its exits are

$$q^{\pm}(R) = m \pm tu, \qquad t = \sqrt{R^2 - a^2}, \qquad C = 2t, \quad \ell = t - h. \tag{9}$$

Here C is the whole sphere-to-sphere segment and ℓ is the added distance at *each* endpoint. The accepted incidence classification gives, on every outward ray, the first subsequent supporting-line meeting at $t_A = w/2$ and the last finite meeting at $t_B = (1+\sqrt{2})w/2$. These are incidence-defined stopping rules; R is not time. Their immediate length consequences are

$$C_A = w, \qquad \ell_A = \left(1 - \frac{\sqrt{2}}{2}\right) w, \qquad C_B = (1 + \sqrt{2})w, \qquad \ell_B = w. \tag{10}$$

Thus one unit appears as a *whole chord* at A and as an *added endpoint extension* at B. Combining the same line equation with $a^2 = w^2/3$ gives

$$\frac{R_A}{w} = \sqrt{\frac{7}{12}}, \qquad \frac{R_B}{w} = \sqrt{\frac{13}{12} + \frac{\sqrt{2}}{2}}. \tag{11}$$

No constant target was used to locate these events. At Original, A and B, forty-two branch labels occupy eighteen distinct outer points; other allowed finite radii have forty-two distinct exits. Splitting at spatial intersections gives the frameworks below, while convexification gives different edge and face counts. [4, Secs. 5, 8–10]

5.1 Why the measurement enclosure reaches A

The earlier measurement construction uses a vertical cylinder of radius $a = w/\sqrt{3}$ and half-height $w/2$. Its two rim circles lie on the sphere

$$R_{\text{cyl}}^2 = a^2 + (w/2)^2 = \frac{7}{12} w^2 = R_A^2. \tag{12}$$

The triangular-prism envelope $K \times [-w/2, w/2]$ has six corners with that same radius and mean zero. The mean-square argument in (6) forces its unique minimum ball; the ball also contains

Table 1: Accepted event geometry at common width w; decimals are rounded. A framework includes interior intersections; its envelope is a separate convex polyhedron.

	Original	Crossing A	Crossing B
Radius R/w	0.613373094	0.763762616	1.338073284
Whole chord C/w	$\sqrt{2}-1$	1	$1+\sqrt{2}$
Per-end addition ℓ/w	0	$1-\sqrt{2}/2$	1
Framework (V, E)	$(18, 21)$	$(36, 63)$	$(54, 105)$
Hull (V, E, F)	$(18, 27, 11)$	$(18, 48, 32)$	$(18, 36, 20)$

the whole cylinder, whose rim circles attain its radius. The six prism corners are precisely the A exits of the three seam lines. This is the accepted cross-source connection between measurement geometry and first line meetings. [10, 12]

The full A framework is larger in different directions: its other exits include $z = \pm w/\sqrt{2}$, and its six equatorial exits have horizontal radius $R_A > a$. Hence the A framework is neither the cylinder nor the prism, despite their shared circumsphere. The original shell is smaller still because it does not attain horizontal radius a and height $w/2$ simultaneously. Shared enclosing radius is the exact correspondence, rather than equality of sets.

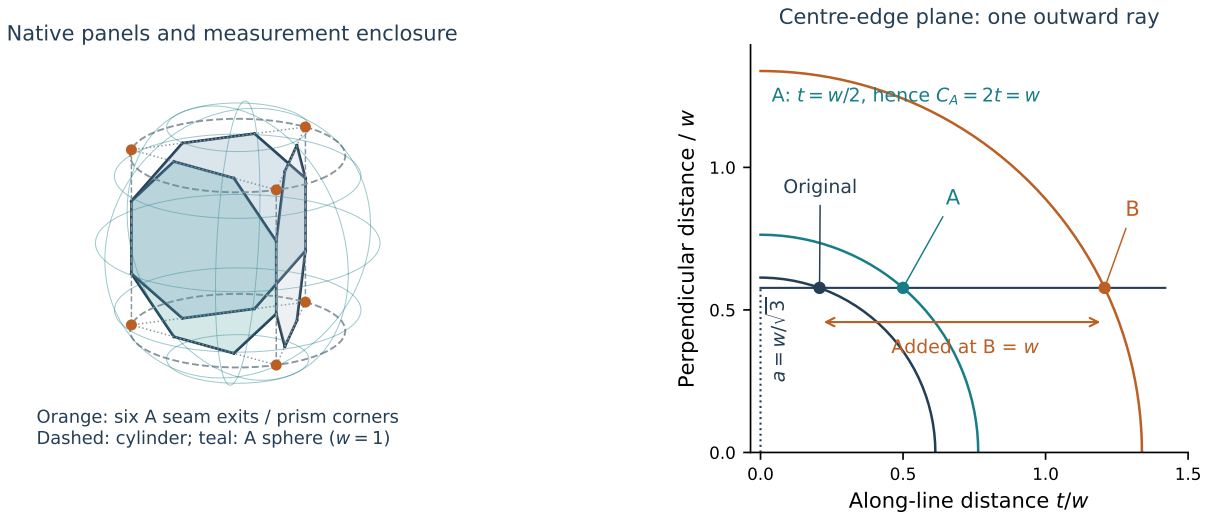


Figure 2: Left: the accepted eighteen-vertex shell at $w = 1$, the measurement cylinder/prism and their A circumsphere. Only the six A seam exits are highlighted; the complete A framework is not depicted as the prism. Right: a plane through the centre and one edge line. Circular sections show Original, A and B at one common scale. The horizontal addition at B is w; the A half-chord is $w/2$. Stored native coordinates and stated formulas were rendered deterministically, without a new geometry solve.

There is another exact bridge to the original shape. The retained TSZ vertex-distance classification gives the shortest native vertex gap as six rim-notch chords with squared length $(3/2-\sqrt{2})w^2$. Since

$$\left(1-\frac{\sqrt{2}}{2}\right)^2 = \frac{3}{2}-\sqrt{2}, \qquad 1-\frac{\sqrt{2}}{2} > 0, \tag{13}$$

their length is exactly ℓ_A. These chords are shorter than the original edges and are not additional welded edges. This equality, explicitly explained in GPT's accepted TSZ review, reuses the distance and event results; it is not a new classification or a claim of global novelty. [11, 12]

6 The historical lock, the exact target and the roles of ten

The author's association of constants with structure preceded this sphere comparison. In particular, T was not selected by inspecting Crossing B. The September 2025 URF manuscript describes a resonance-lock coupling γ, inferred from a run's spectra and adjusted toward

$$\pi e\varphi \approx \gamma^{-1}\zeta(3). \tag{14}$$

Its pages 8–9 call γ emergent and run-dependent. Geometry–Zeta Investigation 01, dated 9 October 2026, then distinguishes the exact fixed target

$$\gamma_* := \frac{\zeta(3)}{\pi e\varphi}, \qquad T := \frac{\zeta(3)}{\gamma_*} = \pi e\varphi \approx 13.817580227176494440. \tag{15}$$

The target is a definition, the historical coupling is a proposed estimator, and Euler–Mascheroni's constant γ_{EM} is a third quantity. The exact target does not retroactively turn every historical coupling estimate into its value. The accepted spherical synthesis preserves this chain in Eqs. (25)–(26), printed p. 18 (physical PDF p. 20), with source distinctions in Appendix E. [13, pp. 8–9] [14, Sec. 1] [4, Sec. 14; App. E]

6.1 An analytic ten, not a selected radius

With $L = \ln\varphi$, the already recorded golden-interval identity is

$$I_2 := \int_0^L x^2 \coth x\, dx = \frac{\zeta(3)}{10}, \qquad \frac{T}{10} = \frac{I_2}{\gamma_*}. \tag{16}$$

The integral is dimensionless; $x^2\coth x \sim x$ at zero, so its lower endpoint is finite. To display the inherited algebra, put $q = \varphi^{-1}$ and use $q + q^2 = 1$. Expanding $\coth x = 1 + 2\sum_{n\geq 1} e^{-2nx}$ and integrating the nonnegative series gives

$$I_2 = \frac{L^3}{3} + \frac{\zeta(3)}{2} - \frac{\mathrm{Li}_3(q^2)}{2} - L\,\mathrm{Li}_2(q^2) - L^2\,\mathrm{Li}_1(q^2). \tag{17}$$

The classical golden values used in the accepted derivation are

$$\begin{aligned} &\mathrm{Li}_1(q^2) = L, && \mathrm{Li}_2(q^2) = \pi^2/15 - L^2, \\ &\mathrm{Li}_3(q^2) = \tfrac{4}{5}\zeta(3) - \tfrac{2\pi^2 L}{15} + \tfrac{2}{3}L^3. \end{aligned} \tag{18}$$

Substitution cancels the L^3 and $\pi^2 L$ terms, leaving $(1/2 - 2/5)\zeta(3) = \zeta(3)/10$. These are inherited analytic identities, with their proof and Hjortnaes/Markov attribution retained in the cited sources, not discoveries of this companion. The second equality in (16) then follows by division by the defined γ_*. It does not independently select a shell width. [14, Sec. 7] [4, App. E, printed pp. 24–25]

The same Geometry–Zeta note separates a spectrum-generated numerator from an adopted normalization. Its auxiliary unit-circumference cycle, for $N \geq 3$, has eigenvalues $\lambda_{N,k} = 4N^2\sin^2(\pi k/N)$. The zero mode is omitted, and for $\alpha > 1/2$ it proves

$$Z_N(\alpha) = \sum_{k=1}^{N-1} \lambda_{N,k}^{-\alpha} \longrightarrow \frac{2\zeta(2\alpha)}{(2\pi)^{2\alpha}}. \tag{19}$$

At $\alpha = 3/2$ the limit is $\zeta(3)/(4\pi^3)$; multiplying by $4\pi^2/(e\varphi)$ gives γ_*. That multiplier inserts the chosen target normalization. The full-cycle spectrum, its zero-mode policy and its inverse moment are explicit additional choices, not a native shell energy law. The note's finite-part energy-type observable is different again. [14, Secs. 3–6]

6.2 Ten as structure, count and convention

Hilmir supplied the June 2025 recursive-toroidal compilation as a reason for choosing ten. Its physical/printed p. 11 explicitly specifies $N = 10$ layers in a model of nested toroidal shells. Pages 7–8 group π, e, φ within a "Stable Harmonic Core" and Apéry's $\zeta(3)$ within "Deep Recursive Excitations"; p. 12 repeats their separation in a later phase classification. These are reported interpretations of that historical study; its simulations and classifications are not revalidated here. The author's present picture extends beyond the layer count: base one and fuller ten suggest a possible extent of an organized whole. His uncertainty about what that extent measures remains explicit. [15, 1]

Several distinct tens therefore coexist. The 10 in (16) is an analytic coefficient; $N = 10$ counts layers. The accepted source ledger also records a historical real-space box-length parameter of ten, without a map to the Tri-Octagon. An assigned observer value such as $R/w = 10$ is yet another convention. Finally, "full size ten" is a proposed structural or tensor extent, whose measure and norm are not specified. The exact recalled aggregate of constants approaching ten has not been recovered. Keeping these roles separate retains the motivation while leaving a concrete place for the missing relation. [4, Sec. 14; App. E] [1, 12]

7 What the retained numerical comparisons establish

Let $b = R_B/w = \sqrt{13/12 + \sqrt{2}/2}$. TSZ examined the explicit counterfactual $w = 10U$, with a target *radius* TU in an arbitrary common length unit U. Then the dimensional radial gap is DU, where

$$D := T - 10b \approx 0.436847382821963879, \tag{20}$$

$$D - 5\gamma_* \approx 0.001873645236788866. \tag{21}$$

The retained outward interval calculation proves $0.0018 < D - 5\gamma_* < 0.0019$. Thus the tested near-equality is not exact; this is not a new search for a replacement coefficient. If T remains an abstract extent with no spatial unit or measure, D is only a scalar subtraction. If the same $w = 10U$ is used for a diameter instead, the B diameter is $20bU$ and it is a different comparison. [16, 12]

There is also a distinction between radial and edge-following additions. Inverting (9) for the assigned target $R/w = T/10$ gives

$$\frac{\ell_{\text{target}}}{w} = \sqrt{(T/10)^2 - 1/3} - \frac{\sqrt{2} - 1}{2}. \tag{22}$$

The retained values are

$$\frac{\ell_{\text{target}} - \ell_B}{w} \approx 0.048250498639102311, \quad \frac{R_{\text{target}} - R_B}{w} \approx 0.043684738282196388. \tag{23}$$

Solving for a target uses that target as input; it does not predict T. The accepted incidence exhaustion already implies that this radius beyond B is a generic forty-two-point observation, with no later finite supporting-line meeting. B remains the event for which $\ell_B = w$. [16, 4]

More generally, the closest distance from an original vertex to the larger spherical surface is $R - \rho$, whereas its literal straight-line extension is ℓ. Along an edge line $dR/dt = t/R < 1$, so for $R > \rho$ the latter is larger. A radial projection $(R/\rho)X$ also changes the component perpendicular to the original edge line from m to $(R/\rho)m$; it is not an extension along that line. The distance convention is therefore part of any proposed relationship, not a choice to be made after a numerical match. [4, Sec. 4.1] [11]

The complement $1 - D \approx 0.563152617178036121$ is well-defined arithmetic after selecting the scalar unit. It does not supply an independently derived energy partition or conservation law. The retained source audit found no native spatial-gap-to-action conversion in its bounded scope; it kept historical scalar analogies, spectral moments and state-dependent intensity/potential diagnostics distinct. The short $\Omega = 1$ Vesica-collapse text likewise uses a proposed R/Q_{env} balance as a symbolic analogy; its R is not identified with our observer radius. It supplies neither a shell-radius derivation nor a calibrated universal energy law. [16, 17]

8 What can now be carried forward

The construction picture and the exact geometry support each other most clearly where the objects are named. The unit-width shell has a unique enclosing sphere even though the author did not originally posit it. First line meetings recover one whole width; last meetings recover one added width at each end. The first event also shares a circumsphere with the earlier measurement enclosure, and its added length equals a pre-existing notch gap. These exact connections require neither a width of ten nor an energy interpretation.

The historical constant chain is also substantial: a proposed lock, a clearly defined target, an exact golden-interval integral and a separately normalized spectral construction. They explain why T, φ and ten were meaningful questions before the present radius comparison. They leave open the map from a proposed total structural extent to a specified length, area, volume or tensor norm. If T and ten measured the same quantity on similar objects, their *linear* scale ratio would be $T/10$ for length, $\sqrt{T/10}$ for area, or $(T/10)^{1/3}$ for volume. That conditional conversion does not make the three event envelopes similar or define a tensor norm. [4, Sec. 14]

Two narrower recollection-led questions remain: whether a specified variable Vesica lock corresponds to the Original radius, and whether the remembered TORMENT alignment near 0.76 shares a defined normalization with $R_A/w \simeq 0.763762616$. No exact parameter match has been supplied, and no application scan is part of this paper. Along with the missing near-ten aggregate, the remembered quantum derivation and the proposed two-to-three-dimensional map, these are identifiable gaps in the source chain, not grounds for declaring that the author had no reasoning or that every future relation is impossible. [1]

Evidence and collaboration. The motivation account and its collaboration clarification are dated 17 October 2026. Historical title-page credits are retained in the bibliography; no older coauthor list is transferred to this companion. Hilmir Frímann Halldórsson is its human author. Codex assisted with source reconstruction, drafting, deterministic diagrams and document preparation. GPT supplied the editorial briefs and internal reviews of the underlying research. The earlier edition received internal manuscript acceptance; Hilmir approved the expanded historical account and publication of v0.2 on 19 October 2026. These author and internal-review roles are not external peer review.

The geometric classifications and TSZ results are retained evidence. GPT freshly reproduced TSZ's scalar checks and 98 metric predicates; its full extrema replay timed out at 360 seconds. Those extrema remain Codex's retained evidence with a reviewed method, not a new independent pass here. This companion performs only source reading, arithmetic/rounding checks of printed scalars and schematic rendering. No old checker, model, application test, dynamics simulation, parameter search or kernel was executed. [12]

References

[1] H. Frímann Halldórsson. *Author intent and source brief*. Retrospective explanations dated 17 October 2026, preserved with separately identified GPT editorial guidance in the GEO-ORIGINS-01 private handoff. Not a contemporaneous 2025 transcript.

[2] H. Frímann Halldórsson. *Author collaboration and narrative placement*. Additive author clarification, 17 October 2026; GPT editorial record for this draft.

[3] H. Frímann Halldórsson. *Exact Geometry of the Folded Tri-Octagon Module*. Paper C, publication v1.0.1, 23 September 2026. Sections 2–5, 7 and 9. Accepted coordinate and normalization authority.

[4] H. Frímann Halldórsson, with Codex preparation and GPT internal review. *Spherical Edge Extensions of the Folded Tri-Octagon: Exact Incidence Events, Natural Terminations, and Convex Envelopes*. Accepted v0.1, 16 October 2026. Sections 2–5, 8–10, 14 and Appendix E. Scientific checkpoint `0997e565f2dd`; full source identities accompany this draft.

[5] H. Frímann Halldórsson, ChatGPT (AI Assistant), Roary (Cat), as credited on the title page. *Vesica Piscis Quantum Wavefunction (VPQW): A Novel Geometric Approach to Quantum Systems*. 28 February 2025. Sections 5.1–5.2 and 6.2, physical/printed pp. 2–3.

[6] H. Frímann Halldórsson, ChatGPT (AI Assistant), Roary (Cat), as credited. *Geometric and Fractal Enhancements to Quantum Correlations within the Vesica Piscis Framework*. 4 March 2025. Figure 1, Section 2 and Section 3; physical/printed pp. 1–3.

[7] H. Frímann Halldórsson, ChatGPT and Roary (cat), as credited. *The Vesica Delay Method: A Geometric-Perceptual Bridge to Coherence*. 2025. Physical/printed p. 1, Eqs. (1)–(2).

[8] *Actual Tri-Octagon Geometry Specification v0.1*. Retained construction record, edge-one normalization; primitive-octagon and local-coordinate sections.

[9] *Actual Tri-Octagon Geometry Specification v0.2*. 20 September 2026. Opening normalization statement and primitive-octagon section; records the author's width-one request.

[10] Codex. *TSZ-01: Implicit and optional spheres*. 17 October 2026. Accepted bounded investigation; cylinder/prism consequences and source trail, reviewed in Ref. [12].

[11] Codex. *TSZ-01: Distance atlas*. 17 October 2026. Accepted retained tables of vertex distances and Original/A/B metrics; definitions use panel width w.

[12] GPT. *TSZ-01 mathematical and source review*. 17 October 2026. Internal acceptance within finite geometry and bounded source-audit scope, Sections 2–5. Includes the extrema-replay limitation and additive historical-ten context.

[13] H. Frímann Halldórsson and ChatGPT 5, as credited. *Unified Recursive Fields: Geometry, Recursion, and Consciousness*. 6 September 2025. Local source `URF_core.pdf`, physical/printed pp. 8–9, Notation and Table of Constants.

[14] *Geometry–Zeta Investigation 01*. 9 October 2026. Sections 1, 3–7. Section 7 credits L. Markov, *Hjortnaes Integrals Involving the Di- and Trilogarithms*, International Journal of Applied Mathematics 37(3), 305–310 (2024), doi:10.12732/ijam.v37i3.2. This companion reuses that accepted attribution, without a fresh external-paper review.

[15] H. Frímann Halldórsson and ChatGPT, as credited. *Topological Complexity in Recursive Toroidal Attractors Seeded by Mathematical Constants*. June 2025 compilation; physical/printed pp. 7–8, 11–12. Actual local copy identified in the source map; added after TSZ's original 43-source inventory.

[16] Codex. *TSZ-01: Zeta, action and residual comparison*. 17 October 2026. Accepted bounded comparison and retained interval result; see Ref. [12] for fresh versus retained verification.

[17] *Vesica Geometry & Toroidal Collapse Interpretation of Stability Number* Ω. Three-page historical local text, no author/date printed in the inspected copy. Sections 1–3, physical/printed pp. 1–2; R/Q_{env} analogy.

[18] H. Frímann Halldórsson. *Expanded retrospective account of the Tri-Octagon's geometric origins*. Author-approved narrative, 19 October 2026; editorial historical expansion authorized for companion v0.2. A recollection rather than a contemporaneous transcript; reproduced in prose in this edition's opening.